

UEC002: PRINCIPLES OF ELECTRONICS

L	T	P	Cr
3	0	2	4.0

Course Objectives: To enhance comprehension capabilities of students through understanding of electronic devices, various logic gates, SOP, POS and their minimization techniques, various logic families and information on different IC's and working of combinational circuits and their applications.

Introductory idea of semiconductors: Formation of P-N junction, energy band diagram, built-in-potential, forward and reverse biased P-N junction, formation of depletion zone. Formation of PNP / NPN junctions, energy band diagram.

Diodes and Diode Circuits: V-I characteristics, Zener breakdown, Avalanche breakdown and its reverse characteristics; Junction capacitance and Varactor diode. Simple diode circuits, load line, linear piecewise model; Rectifier circuits: half wave, full wave, PIV, DC voltage and current, ripple factor, efficiency, idea of regulation.

Transistors and transistor circuits: Transistor mechanism and principle of transistors, CE, CB, CC configuration, transistor characteristics: cut-off active and saturation mode Concept of Field Effect Transistors (channel width modulation), Gate isolation types, JFET Structure and characteristics, MOSFET Structure and characteristics, depletion and enhancement type; CS, CG, CD configurations; CMOS: Basic Principles.

Operational amplifier basics: Introduction to integrated circuits, operational amplifier and its terminal properties; Application of operational amplifier; inverting and non-inverting mode of operation, Proportional, Integral, Derivative circuits.

Basic ideas of Digital electronics: Basic idea of switching circuit, Realization of Logic gates, multiplexers and demultiplexers, Flip flop, Registers and Counters.

Laboratory Work:

1. Familiarization with CRO, DSO, electronic components(Resistor, capacitor, inductor, LED, seven segment display, diodes, transistor, logic gates, integrated circuits)

2. Experiments cover studying VI characteristics of pn and zener diode, half wave and full wave rectifier, design clipper circuits, switching characteristics of BJT
3. Verification of De-Morgan laws, logic gates; implementation of Boolean functions, Half Adder, Full Adder using Multiplexers and Encoders.
4. To study and analyze the working of inverting and non-inverting amplifier, amplifier as summer and average circuits.
5. Simulation of above analog and digital circuits using Multisim software

Course Learning Outcomes (CLOs) / Course Objectives (COs):

The student will be able to:

1. Demonstrate the use of semiconductor diodes in various applications.
2. Discuss and explain the working of transistors, their configurations and applications.
3. Design and implement combinational and sequential circuits.
4. Reduce Boolean expressions and implement them using Logic ICs.
5. Discuss and analyze the concept, working and applications of FETs, Operational Amplifiers and Oscillators.

Text Books:

1. Adel S. Sedra and Kenneth Carless Smith, Microelectronics Circuits, Oxford University Press.
2. Jacob Millman, Christos Halkias, Chetan Parikh, Millman's Integrated Electronics, McGraw Hill Education.

Reference Books:

1. Robert L. Boylestad, Louis Nashelsky, Electronic Devices & Circuit Theory 11th ed.
2. D Schilling C Belove T Apelewicz R Saccardi, Electronics Circuits: Discrete & Integrated.